

xchan: An R Package for Channel Cross-Section Analysis

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Summary

xchan is an R package that helps researchers analyze river channels by creating and manipulating cross-sectional data. River channels can be described in two ways: as horizontal lines across the channel (planimetric cross-sections) and as elevation profiles along those lines (profile cross-sections). The package allows users to generate these cross-sections from various data sources including river bank outlines, digital elevation models, or simple width specifications. Once created, users can model bank widening and erosion and summarize channel geometry (for example width, gradient, and elevations). A channel axis can be traced along the cross sections, and other derived geometries (such as a thalweg line when profiles are available) support visualization and analysis. The package integrates with existing R spatial analysis tools, making it easy to incorporate into larger research workflows for studying river processes, erosion, and channel evolution in a programmatic way.

Statement of Need

River channel analysis requires handling both planimetric cross-sections (horizontal line segments extending bank-to-bank) and profile cross-sections (elevation data along those segments). Current approaches often require manual GIS workflows or separate tools that don't integrate well with the R ecosystem or automation. Existing R packages for spatial analysis (e.g., **sf** (Pebesma 2018), **terra** (Hijmans 2023)) provide excellent tools for individual operations but lack specialized functionality for channel cross-section analysis.

The advent of remote sensing technologies, particularly airborne LiDAR and photogrammetry, has provided the ability to capture high-resolution topographic data over large spatial extents (Milan et al. 2011; Westaway et al. 2003). These technologies enable the generation of digital elevation models (DEMs) that can be used to extract detailed channel and floodplain geometry information. However, the challenge lies in efficiently processing these data to extract meaningful cross-sectional information and integrate it with other spatial datasets.

Current approaches to channel analysis rely on manual GIS workflows using software such as ArcGIS or QGIS, where users must manually digitize cross-sections, extract elevation profiles, and perform geometric calculations through point-and-click interfaces. These workflows are time-consuming, error-prone, and difficult to reproduce. While R packages like **sf** and **terra** provide excellent tools for spatial data manipulation and raster processing, they lack specialized functionality for channel cross-section analysis. Researchers must cobble together multiple packages and custom scripts to achieve what should be a unified workflow: generating cross-sections from banklines, sampling DEMs for elevation data, and analyzing channel geometry properties.

This gap is particularly evident in the context of fluvial geomorphology research, where researchers need to analyze channel geometry changes over time, assess erosion and deposition patterns, and model channel evolution processes (Church 2006; Eaton et al. 2010). The **xchan** package addresses this gap by providing a unified framework for channel cross-section analysis that integrates seamlessly with the existing R spatial analysis ecosystem.

Key Features

The package provides a unified data structure for channel objects that integrates planimetric and profile cross-sections. Plan-view cross-sections extend from bank to bank, while profile cross-sections are elevation profiles along those lines. To enable informed erosion modeling into the floodplain, profile cross-sections can extend beyond the banks into the surrounding topography.

Data Structures

The package deals with two main object types:

- **Channels:** The primary objects, comprised of cross-sections
- **Profile cross-sections:** Specialized geometry objects requiring custom handling

Plan-view cross-sections are represented as simple line segments (using `sf` geometries), avoiding unnecessary complexity and keeping the package lightweight.

Channel Objects

Channels are implemented as data frames (or tibbles (Müller and Wickham 2020) when available), with plan and profile cross-sections stored as columns. Additional columns can represent channel properties at each cross-section, following the design philosophy of the `sf` package where spatial objects are just enhanced data frames.

Profile Cross-Sections

Profile cross-sections are specified as coordinate matrices with distance along the profile in the first column and elevation in the second. They encode bank locations and thalweg points (potentially multiple thalwegs for complex channels).

Design Philosophy

The package prioritizes human readability and consistency. Functions are prefixed with `xt_` (for “cross-section type”) to avoid conflicts and improve code clarity, following the pattern established by the `sf` package. The `units` package ensures consistent dimensional analysis throughout. The gradient calculation function draws inspiration from the `slider` package’s sliding window approach for time series analysis (Vaughan 2024).

Function Factory Architecture

A core architectural decision is the consistent use of function factories for configurable operations. This pattern elegantly solves the tension between API simplicity and extensibility that plagues many scientific software packages. Rather than creating separate functions for each algorithm variant (e.g., `xt_widen_linear()`, `xt_widen_spline()`, `xt_widen_custom()`), the package uses factory functions that return specialized algorithms while maintaining a unified interface.

This design provides several key advantages:

Consistent User Experience: Configurable behaviours such as erosion allocation and elevation references follow the same factory-based pattern, so users can reuse ideas across widening, erosion metrics, and related workflows.

Parameter Capture: Factory functions elegantly capture configuration parameters in closures, avoiding the need for complex parameter passing or global state.

Extensibility Without API Bloat: New algorithms can be added without changing the core function signatures. Users can create custom implementations that integrate seamlessly with the existing API.

R Ecosystem Integration: This pattern is widely used in the R ecosystem (e.g., `ggplot2` geoms, `dplyr` verbs, `scales` transforms), making the package feel familiar to R users.

The package implements **specification factories** for operations that need flexible, composable parameters:

- **Side Factories** create functions that specify erosion allocation during channel widening, allowing users to define how widening or erosion is distributed between left and right banks. Built-in factories handle common cases (e.g., "left", "right", "both"), and custom functions can vary allocation along the channel.
- **Elevation Factories** create functions for flexible elevation specification (thalweg, banks, beyond-bank topography), so analyses can reference consistent vertical datums across cross sections.

These factories return objects that can be passed as arguments and composed with core geometry functions, avoiding a proliferation of similarly named top-level functions while keeping call sites readable.

This architectural approach transforms what could be a collection of disparate functions into a cohesive, extensible system that grows with user needs rather than requiring API changes.

Channel Operations

Channel Widening

Channel widening simulates bank erosion processes. For plan-view cross-sections, widening simply extends the line segments. For profile cross-sections, the approach preserves the existing profile while widening the channel from the thalweg outward. This means the channel bottom (thalweg) becomes wider, with the river section between the bank and thalweg shifting laterally, and corresponding beyond-bank topography being removed.

Derived Geometries

The package provides functions to generate derived geometries from channel objects. For example, a channel axis can be traced by connecting the midpoints of planimetric cross-sections (`xt_trace_centerline()`), and where profile cross-sections are present, a thalweg line can be assembled in plan view (`xt_trace_thalwegline()`).

Example Usage

```
library(xchan)
library(sf)

# Channel from nominal widths at each cross section (plan geometry only)
channel <- xt_as_channel(c(10, 12, 8, 15, 11, 9))

# After attaching a DEM raster `dem`, build profiles along existing plan sections:
# channel <- xt_generate_profile(
#   channel,
#   dem,
#   extent_multiplier = 2,
#   sample_n = 100
# )

# Widen with asymmetric erosion allocation (must pass dw or dv by name)
channel_widened <- xt_widen(channel, dw = 10, side = side_left(0.75))

# Derived centreline in plan view
centerline <- xt_trace_centerline(channel)

# Summaries
```

```
widths <- xt_width(channel_widened)
gradients <- xt_gradient(channel_widened)
```

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